

# LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

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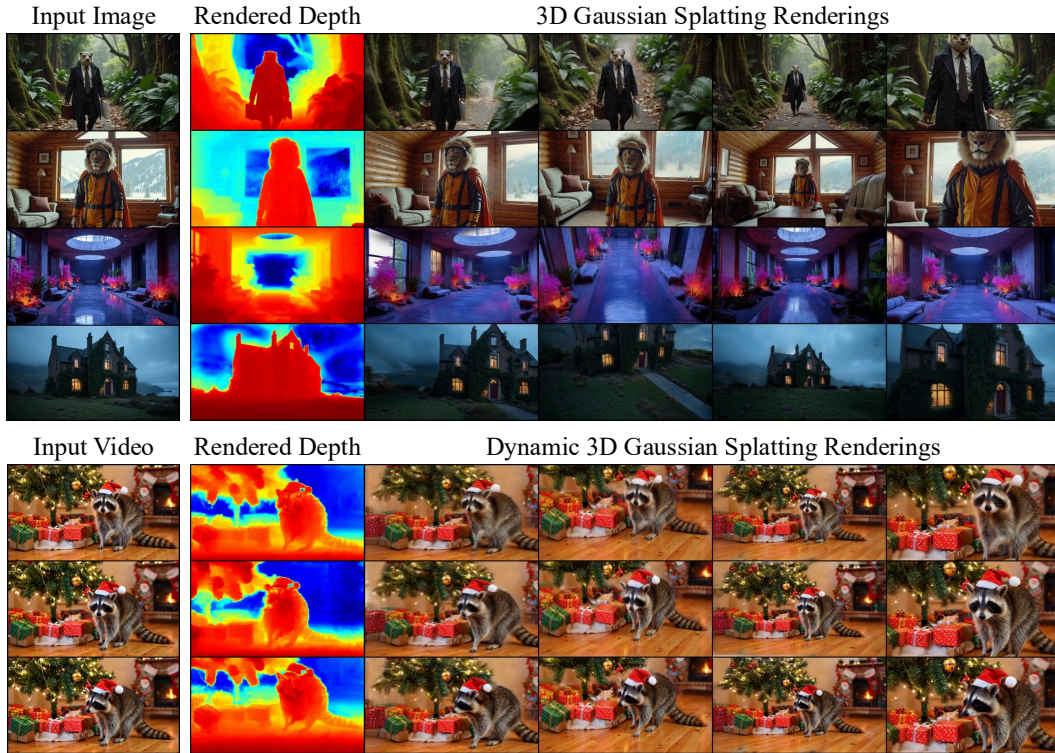


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** Conditioned on a single image (top), *Lyra* infers a 3D Gaussian Splatting (3DGS) representation in a single feed-forward pass by self-distilling a video diffusion model, requiring no real-world multi-view data. Conditioned on a video (bottom), *Lyra* infers a dynamic 3DGS that supports independent control over time (rows) and viewpoint (columns).

## ABSTRACT

The synthesis of virtual environments is a foundational capability for applications spanning gaming and physical-AI domains such as robotics, autonomous driving, and industrial AI. Existing learning-based 3D reconstruction pipelines are predicated on the availability of captured real-world multi-view observations, a prerequisite that is frequently violated in practice. Video diffusion models have recently exhibited strong generative priors over natural scenes, yet their strictly 2D output space precludes deployment in simulation settings that require an agent to navigate and interact with the environment. In this paper, we formulate a self-distillation framework that transfers the 3D structure implicitly encoded by video diffusion models into an explicit 3D Gaussian Splatting (3DGS) representation, thereby removing the dependence on multi-view training data. Concretely, we augment the standard RGB decoder with a parallel 3DGS decoder whose renderings are supervised directly against the RGB decoder’s outputs. Under this construction, the 3DGS decoder is optimized exclusively from synthetic supervision generated by video diffusion models. At inference, the model synthesizes 3D scenes conditioned

on either a text prompt or a single image, supporting real-time rendering. The framework generalizes to dynamic 3D scene generation conditioned on a monocular input video. Empirically, the proposed framework attains state-of-the-art performance on both static and dynamic 3D scene generation.

## 1 INTRODUCTION

Synthesizing high-quality 3D environments at scale remains a long-standing problem in computer vision and graphics, underpinning applications in film production, VR/AR, and physical AI with closed-loop simulation. These applications impose a common requirement: an explicit 3D representation that admits real-time rendering, physical interaction, and geometrically consistent multi-view synthesis.

Neural 3D reconstruction methods (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) recover such representations from posed image sets. However, their dependence on accurate camera poses and high-quality imagery imposes a substantial scalability constraint. This constraint is exacerbated in the dynamic setting, which typically mandates synchronized multi-camera rigs (Li et al., 2022). Reconstruction is furthermore intrinsically restricted to observed content and cannot extrapolate beyond the supplied views. Feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c) were introduced to relax these constraints, but they are in turn bottlenecked by the scarcity of diverse, large-scale 3D training corpora, which manifests as poor out-of-domain generalization.

Video diffusion models (Cosmos, 2025; Wan, 2025) constitute a promising alternative. Optimized on internet-scale video corpora, they attain high fidelity and generalize across heterogeneous environments. By fitting to real-world videos spanning diverse camera trajectories, ranging from handheld motion to cinematic panning and drone footage, these models implicitly encode cues about the underlying 3D scene without any multi-view training supervision. In contrast to reconstruction approaches (Charatan et al., 2024), generative models can additionally synthesize plausible content in regions unobserved in the input frames. Nonetheless, video diffusion models emit only 2D frames and expose no explicit 3D representation, which precludes their use in simulation and in downstream tasks that require geometric consistency, long-horizon coherence, and physical interaction. Encouragingly, recent work (Wang et al., 2024e; Ren et al., 2025) demonstrates that video models can be conditioned on explicit camera control using comparatively small posed datasets. The resulting capacity to generate posed image sequences renders them a versatile instrument that can serve simultaneously as input and as supervision for 3D reconstruction models.

Motivated by this observation, we unify the two paradigms, reconstruction and generation, under a formulation we term *generative 3D scene reconstruction*. We present *Lyra*, a method that synthesizes an explicit 3D environment from the latent representation of a video diffusion model in a single forward pass. Central to *Lyra* is a self-distillation framework (Fig. 2) in which a 3D Gaussian Splatting (3DGS) decoder operates directly on the latent space of a video diffusion model. Given a single input image or video, we sample a camera trajectory as the conditioning signal, denoise the induced video latent, and decode it along two parallel branches. The first branch applies the pre-trained RGB decoder to synthesize a video sequence; the second is the proposed 3DGS decoder, which emits an explicit 3D representation. Jointly, the two branches instantiate a self-distillation objective in which the RGB branch (teacher) supervises the 3DGS branch (student). To broaden viewpoint coverage, we sample multiple camera trajectories, generate the corresponding video latents, and train the 3DGS decoder to fuse information across them while suppressing long-horizon and multi-view inconsistencies.

This generative reconstruction framework confers three principal advantages: **(i)** it enables synthesis of large-scale synthetic environments spanning diverse scenarios directly from video diffusion models, obviating real-world multi-view capture; **(ii)** by operating in the latent space of a video model, it processes multiple views efficiently and avoids the memory overhead incurred by pixel-space feed-forward reconstruction; and **(iii)** its explicit 3DGS output enforces geometric consistency, yielding representations directly consumable by downstream tasks such as physical simulation. Consequently, at inference, *Lyra* produces high-quality 3D Gaussian scenes from monocular input that support real-time rendering, *without any additional optimization or post-processing*.

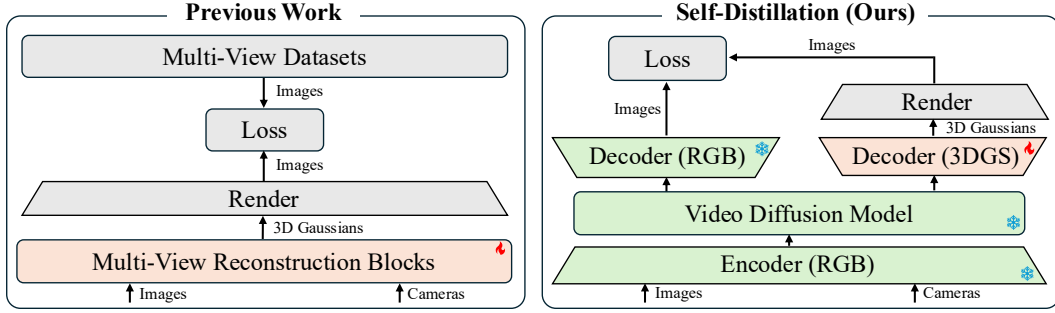


Figure 2: **Self-distillation framework of *Lyra***. Prior work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks on real-world datasets of limited diversity (Zhou et al., 2018; Ling et al., 2024b). In contrast, we formulate a self-distillation framework (right) for generative 3D scene reconstruction, in which the RGB decoder output of a pre-trained camera-controlled video diffusion model (teacher) supervises the renderings of the 3DGS decoder (student). Because the video model is pre-trained on diverse 2D video data, it furnishes correspondingly diverse multi-view supervision.

We further generalize the self-distillation framework to dynamic 4D generation from monocular video. In this regime, the video model (teacher) supplies spatiotemporal supervision, while the student is trained to emit time-conditioned 3DGS representations that support novel-view synthesis of dynamic scenes.

In summary, this work contributes the following:

- We formulate a self-distillation framework that trains a 3DGS student decoder using a pre-trained camera-controlled video diffusion model as the teacher, eliminating the requirement for captured multi-view real-world data.
- We show that the framework extends to dynamic scenes, supporting 4D reconstruction from a monocular video.
- We demonstrate that the resulting model generalizes across diverse scenes, attaining state-of-the-art results on single-image 3D scene generation and single-video 4D scene generation.

## 2 RELATED WORK

**Multi-view image generation.** Early multi-view image generation methods (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) are largely confined to object-centric scenes without background. Subsequent methods (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) scale multi-view image generators to the scene level. CAT3D (Gao et al., 2024b) adapts a pre-trained image diffusion model to multi-view image generation conditioned on an arbitrary generated or real image. A subsequent stage reconstructs the generated multi-view images into an explicit 3D representation, e.g., NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023). These formulations have further been extended to dynamic 3D scenes observed from multiple viewpoints via a space-time diffusion model (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). Although these methods yield compelling results, grounding the generations in an explicit 3D representation entails a costly per-scene optimization stage whose cost is not amortized across scenes. Rather than generating multi-view images and subsequently reconstructing via optimization, we target the *direct* synthesis of a 3D scene from text or a single image.

**Camera-conditioned video generation.** Substantial progress has been made in fine-tuning video diffusion models for 3D camera control. MotionCtrl (Wang et al., 2024e) introduced camera control by conditioning pre-trained video models on camera poses. Subsequent methods (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) encode cameras as Plücker coordinates to enable pixel-wise conditioning. A separate line of work (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) investigates training-free camera control of video diffusion models. More recently, SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) achieved high-quality camera-controlled video synthesis by leveraging large-scale synthetic training data. While these methods produce compelling results, their outputs are not grounded in an explicit 3D representation. Our work is orthogonal: we address feed-forward 3D generation from camera-controlled video models, which permits novel-viewpoint rendering *after* generation. Concretely, we adopt GEN3C (Ren et al., 2025), a recent camera-controlled video diffusion model, as the teacher within a self-distillation framework for 3D reconstruction.

**Feed-forward 3D models.** Early feed-forward 3D reconstruction methods were largely restricted to object-centric scenes without background, directly regressing a NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) from text or a single image. More recent methods (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) target scene-level 3D reconstruction, typically regressing per-pixel 3D Gaussians from one or more input images. However, most scene-level reconstruction methods remain confined to the training distribution, e.g., RealEstate10K (Zhou et al., 2018), and generalize poorly to generated scenes. A recent thread extends feed-forward 3D reconstruction models toward scene generation. Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder that jointly generates pointmaps and multi-view images, which are then consumed for feed-forward 3D reconstruction. Most closely related to our work, Wonderland (Liang et al., 2025a) employs a camera-controlled video model to regress 3D Gaussians through a feed-forward network. In contrast to these methods, we treat a camera-controlled video model as a teacher within a self-distillation framework to train our student 3D model *without any real-world multi-view data*. Importantly, we further present a minimal extension of the framework to feed-forward 4D scene generation, a task not previously addressed.

### 3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

The central idea is to distill the 3D structure implicitly encoded by video diffusion models into an explicit 3DGS decoder that produces high-quality 3D representations. To this end, we construct a teacher–student framework in which the video diffusion model (teacher) generates RGB videos that supervise the 3DGS decoder (student), with both operating in a shared latent space, as illustrated in Fig. 2. This section specifies the video diffusion backbone (Sec. 3.1) and its role in self-distillation for the 3DGS decoder (Sec. 3.2). We then motivate the design of the 3DGS decoder in Sec. 4 and delineate the minimal modifications required to adapt the pipeline to dynamic 3D scenes in Sec. 5.

#### 3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Because the 3DGS decoder is supervised exclusively by the camera-controlled video diffusion model, its accuracy is contingent on the 3D consistency of that model. We therefore instantiate our approach on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model. We summarize the salient design choices of GEN3C below.

**Spatiotemporal 3D cache.** To improve temporal consistency and the precision of camera control, GEN3C constructs a spatiotemporal 3D cache  $\{\mathbf{P}^{t,v}\}$  from the input image(s) or video, where each  $\mathbf{P}^{t,v}$  is a colored point cloud obtained by unprojecting the estimated depth (Wang et al., 2024a) of an RGB image observed from camera viewpoint  $v$  at time  $t$ . The cache is organized as an  $L \times V$  array, where  $L$  denotes the temporal length (number of frames) and  $V$  denotes the number of camera views.

**Rendering and structured guidance.** To exploit this cache, GEN3C renders each point cloud from arbitrary specified camera poses, yielding RGB images  $\mathbf{I}^{t,v}$  and disocclusion masks  $\mathbf{M}^{t,v}$  through  $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$ , where the rendering operator  $\mathcal{R}$  projects the 3D point cloud  $\mathbf{P}^{t,v}$  onto the 2D image plane according to the camera pose  $\mathbf{C}^t$  at time  $t$ . Given a sequence of camera poses  $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$ , rendering all cache elements produces  $V$  videos of length  $L$ , which are stacked into image sequences  $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$  and mask sequences  $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$  per view  $v$ . The disocclusion masks demarcate the regions to be inpainted by the video diffusion model. Collectively, these renderings constitute structured visual guidance for the subsequent video generation stage.

**Video variational autoencoder (VAE).** Diffusion-based video generation models typically employ a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) to compress videos into a lower-dimensional latent space, thereby reducing the cost of training and inference. Given an RGB video  $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$ , a pre-trained VAE encoder  $\mathcal{E}$  maps the video to the latent space, i.e.,  $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$ . Both training and inference of the diffusion model are conducted in this latent space. The output video  $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$  is recovered by a pre-trained VAE decoder  $\mathcal{D}_{rgb}$ . We adopt the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model, for which the latent channel dimension is  $C = 16$ , the temporal dimension is  $L' = (L - 1)/\tau + 1$ , and the spatial dimensions are  $h = H/\sigma$  and  $w = W/\sigma$ , with temporal compression factor  $\tau = 8$  and spatial compression factor  $\sigma = 8$ .

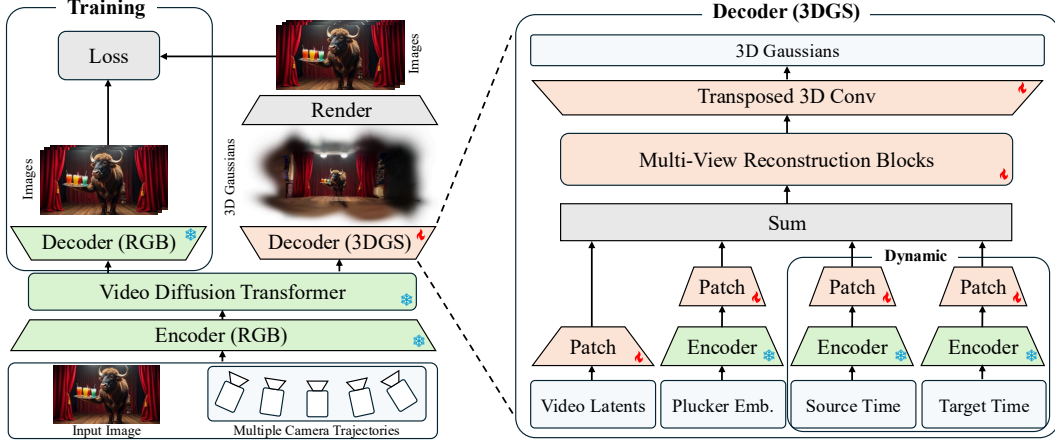


Figure 4: **3D Generative Reconstruction Framework.** The pipeline is built upon a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large-scale data. We train a 3D Gaussian Splatting (3DGS) decoder by aligning the 2D image renderings of the generated 3DGS scenes with the RGB-decoded generations of the pre-trained video model. Only the 3DGS decoder is trained; the pre-trained autoencoder and diffusion model remain frozen. At inference, the 3DGS decoder is used in isolation, with the RGB decoder no longer required. Time conditioning within the 3DGS decoder enables a direct extension from static to dynamic 3D scene generation.

### 3.2 SELF-DISTILLATION

We train the 3DGS decoder  $\mathcal{D}_s$  as the student in a teacher–student paradigm, with a camera-controlled video diffusion model  $\mathcal{V}$  acting as the teacher. To obtain diverse supervision, we curate a large-scale collection of text prompts with large language models (OpenAI, 2025; Bai et al., 2023), synthesize images  $I$  with an image diffusion model (Black Forest Labs, 2024), and expand each image into multi-view sequences using GEN3C (Ren et al., 2025).

**Teacher–student setup.** Given an input image  $I$  and a sampled camera trajectory  $\{\mathbf{C}^t\}_{t=1}^L$ , the video diffusion model  $\mathcal{V}$  produces a denoised video latent  $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$ . The latent  $\mathbf{z}$  is decoded along two branches: the pre-trained RGB decoder  $\mathcal{D}_{rgb}$  yields video frames  $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$ , while the 3DGS decoder  $\mathcal{D}_s$  emits explicit 3D Gaussians  $\mathbf{G}$ . The rendered student views  $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$  are constrained to match the teacher frames  $\mathbf{I}_{\mathcal{D}_{rgb}}$ , thereby closing the self-distillation loop.

**Multi-trajectory supervision.** To enlarge the viewpoint coverage of a given input image, we sample  $V = 6$  camera trajectories per image, as shown in Fig. 3. For each trajectory  $v$ , we construct a spatiotemporal cache  $\{\mathbf{P}^{t,v}\}_{t=1}^L$  together with  $L = 121$  camera poses  $\{\mathbf{C}^{t,v}\}_{t=1}^L$ . Propagating these caches through the video diffusion model  $\mathcal{V}$  produces latents  $\mathbf{z}^v$ , which the teacher decodes into RGB frames  $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$ . The 3DGS decoder  $\mathcal{D}_s$  is trained to fuse the multiple latents  $\mathbf{z}^v$  into a single coherent set of Gaussians  $\mathbf{G}$  while inpainting disoccluded regions. Under this self-distillation design,  $\mathcal{D}_s$  is optimized entirely from the synthetic supervision furnished by  $\mathcal{V}$ . Operating in latent space permits efficient aggregation across trajectories, whereas the explicit Gaussian output  $\mathbf{G}$  enforces geometric consistency and enables downstream simulation and real-time rendering.

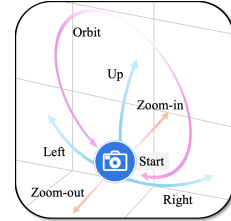


Figure 3: The six camera trajectories sampled per input to maximize viewpoint coverage.

## 4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

The feed-forward 3DGS decoder  $\mathcal{D}_s$  is designed to map the synthesized multi-view latents produced by the video model  $\mathcal{V}$  to an explicit 3D representation renderable from arbitrary viewpoints. We adopt 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) on account of its explicit parameterization and fast rasterization-based rendering. The detailed decoder architecture is depicted in Fig. 4 (right).

#### 4.1 3DGS DECODER

**Scaling Latent-Based 3D Reconstruction.** Prior feed-forward reconstruction frameworks regress one 3D Gaussian per input pixel and are therefore bounded by the number and resolution of input views they can accommodate. For instance, GS-LRM (Zhang et al., 2024e) operates on 2–4 images at  $512 \times 512$  resolution, and AnySplat (Jiang et al., 2025a) is trained on 24 images at  $448 \times 448$  resolution. By contrast, our video model synthesizes  $V \times L = 6 \times 121 = 726$  input views at  $704 \times 1280$  resolution, an input scale that exceeds the capacity of existing methods and the available GPU memory. The limiting factor is the attention mechanism over visual tokens, whose memory and compute costs scale with the number of pixels. To circumvent this limitation, we forgo scaling in pixel space and instead operate directly on the compressed video latent space produced by the camera-controlled video diffusion model. The multi-view video latents are denoted  $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$ .

**Architecture.** The 3DGS decoder  $\mathcal{D}_s$  first projects its inputs to the hidden dimension of the main reconstruction blocks and regresses per-pixel 3D Gaussian features  $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$ . The inputs comprise the multi-view video latents  $\mathbf{Z}$  and the encoded Plücker embeddings  $\mathbf{E}$ , both of latent dimensionality  $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$ . Each input is patchified to the hidden dimension of the reconstruction blocks. No auxiliary visual encoder is required; a single spatial  $2 \times 2$  patchification layer (Dosovitskiy et al., 2021) suffices to convert the video latents into flattened tokens for the reconstruction network. The element-wise sum of the two inputs is then processed by the reconstruction blocks. Following the design of Long-LRM (Ziwen et al., 2024), each block comprises one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers; the block is instantiated twice, yielding 16 layers with a hidden width of 512. Finally, a transposed 3D convolution maps the hidden representation to the 14 Gaussian channels: 3D position  $(x, y, z)$ , scale  $(s_x, s_y, s_z)$ , rotation quaternion  $(q_w, q_x, q_y, q_z)$ , opacity  $\alpha$ , and color  $(r, g, b)$ . Formally, the static decoder is written  $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$ .

**Plücker embeddings.** Raw Plücker embeddings are computed from the camera poses  $\{\mathbf{C}^{t,v}\}$  and intrinsics as  $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$ . We reuse the RGB encoder  $\mathcal{E}$  of the pre-trained VAE to embed the Plücker coordinates. Concretely, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins with  $\mathcal{E}$ , then concatenate the resulting latents along the channel axis to obtain  $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$ . A lightweight MLP maps  $\mathbf{E}^{\text{enc}}$  to  $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$ , halving the channel dimension to align with the video latent  $\mathbf{Z}$ .

#### 4.2 LOSS FUNCTION

**Image-based supervision.** The 3DGS decoder is supervised with an image-space reconstruction objective decomposed into a mean-squared-error term  $\mathcal{L}_{mse}$  and a perceptual LPIPS term  $\mathcal{L}_{lips}$  (Zhang et al., 2018), the latter using VGG (Simonyan & Zisserman, 2014) as the feature extractor.

**Depth supervision.** We empirically observe that a decoder trained with the RGB objective alone tends to produce degenerate, near-planar geometry. To counteract this, we additionally supervise the rendered depth maps against the temporally consistent video depth estimated by the off-the-shelf system ViPE (Huang et al., 2025), using the scale-invariant depth loss  $\mathcal{L}_{depth}$  of Long-LRM (Ziwen et al., 2024).

**Opacity-based pruning.** Following Long-LRM (Ziwen et al., 2024), we impose an  $L_1$  regularizer  $\mathcal{L}_{opacity}$  on the Gaussian opacities and discard the 80% of Gaussians with the lowest opacity.

**Total loss.** The total objective is the weighted sum of the individual terms with coefficients  $\lambda_i$ ,

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lips} \mathcal{L}_{lips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

with  $\lambda_{mse} = 1.0$ ,  $\lambda_{lips} = 0.5$ ,  $\lambda_{depth} = 0.05$ , and  $\lambda_{opacity} = 0.1$ .

## 5 EXTENSION TO DYNAMIC 3D SCENES

The proposed approach extends to dynamic 3D scenes with only minimal modifications, detailed below.

**Self-distillation for dynamic scenes.** The dynamic setup mirrors its static counterpart. In place of a single image, the video model is conditioned on a single video of length  $L$  together with its camera

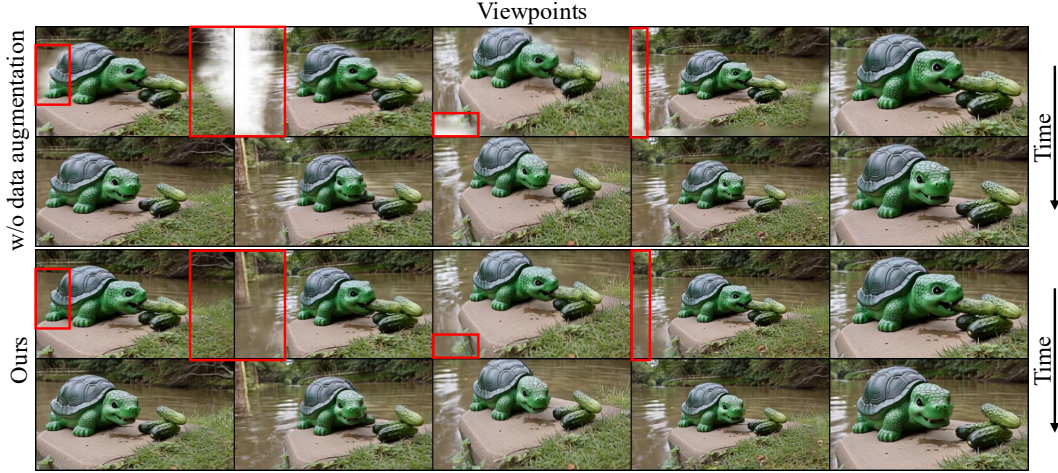


Figure 5: **Dynamic data augmentation.** Naive supervision of the time-aware 3DGS decoder induces artifacts in the generated 3D Gaussians: early timesteps observed from extreme poses exhibit low opacity in regions uncovered by the supervision signal. To remedy this, we augment the supervision data with a motion-reversed video so that each timestep is observed under full spatial coverage, thereby eliminating the low-opacity artifacts at early timesteps.

poses, and generates multi-view video latents that capture the same underlying motion. For video inputs we adopt a protocol analogous to the static case: text prompts are sampled with large language models, and videos are then synthesized with video diffusion models (Cosmos, 2025; Wan, 2025). The generated videos are subsequently annotated with camera poses and depth maps by ViPE (Huang et al., 2025).

**Dynamic 3DGS decoder architecture.** Following the bullet-time formulation of Liang et al. (2025b), we design the dynamic 3DGS decoder  $\mathcal{D}_s$  to regress time-dependent 3D Gaussians for every video frame. The architecture of  $\mathcal{D}_s$  largely coincides with that of  $\mathcal{D}_s$ , the difference being that the input of  $\mathcal{D}_s$  is augmented with encoded source and target time embeddings  $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$ . The raw (unencoded) source times are assigned per input frame, forming  $\mathbf{T}^{\text{src, raw}} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$ , while the target time is  $\mathbf{T}^{\text{tgt, raw}} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$ . Both tensors are spatially broadcast and concatenated with a 2-dimensional sinusoidal embedding to raise the channel dimension to 3, then mapped by the RGB encoder  $\mathcal{E}$  to latent dimensions  $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$ . The target time is replicated along the latent temporal axis to match  $L'$ . At training time, a target timestep is sampled uniformly at random, and the corresponding 3D Gaussians are decoded and supervised across all trajectories produced by dynamic data augmentation. We fine-tune  $\mathcal{D}_d$  from a pre-trained  $\mathcal{D}_s$ , initializing the patchification layers of  $\mathbf{T}^{\text{src}}$  and  $\mathbf{T}^{\text{tgt}}$  to zero.  $\mathbf{T}^{\text{src}}$  and  $\mathbf{T}^{\text{tgt}}$  are added directly to the sum of  $\mathbf{Z}$  and  $\mathbf{E}$ . Formally, the dynamic decoder is defined as  $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}})$ .

**Dynamic data augmentation.** For static scenes, where time is not a factor, frames from all timesteps are valid supervision for the 3DGS renderings. For dynamic scenes, however, the 3DGS varies over time, so only frames sharing the target timestamp constitute valid supervision. This restriction admits a degenerate solution in which the 3DGS at a given timestep disregards information from frames at other timesteps (see Fig. 5).

To preclude this failure mode, we introduce a dynamic data augmentation strategy that balances supervision across timesteps. Early frames are inherently associated with viewpoints near the input image, whereas later frames correspond to more distant viewpoints. To offset this imbalance, we augment the training data with a pair of supervision views per timestep, one near and one far. Concretely, we reverse the input video in frame order and pass it through the video model  $\mathcal{V}$ , obtaining six additional multi-view sequences. Reversing these sequences back to the original motion order yields six trajectories that progress inward (from distant viewpoints toward the input). Together with the original six outward trajectories, this produces 12 supervision views per timestep. Crucially, the augmentation is confined to training: the reversed trajectories serve solely as supplementary supervision that prevents collapse under pruning artifacts and are not required at inference.

Table 1: **State-of-the-art comparisons.** We benchmark our method against prior work on single-image-to-3D generation across the RealEstate10K, DL3DV, and Tanks-and-Temples datasets.

| Method      | RealEstate10K   |                 |                    | DL3DV           |                 |                    | Tanks-and-Temples |                 |                    |
|-------------|-----------------|-----------------|--------------------|-----------------|-----------------|--------------------|-------------------|-----------------|--------------------|
|             | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ | PSNR $\uparrow$   | SSIM $\uparrow$ | LPIPS $\downarrow$ |
| ZeroNVS     | 13.01           | 0.378           | 0.448              | 13.35           | 0.339           | 0.465              | 12.94             | 0.325           | 0.470              |
| ViewCrafter | 16.84           | 0.514           | 0.341              | 15.53           | 0.525           | 0.352              | 14.93             | 0.483           | 0.384              |
| Wonderland  | 17.15           | 0.550           | 0.292              | 16.64           | 0.574           | 0.325              | 15.90             | 0.510           | 0.344              |
| Bolt3D      | 21.54           | 0.747           | 0.234              | -               | -               | -                  | -                 | -               | -                  |
| Ours        | <b>21.79</b>    | <b>0.752</b>    | <b>0.219</b>       | <b>20.09</b>    | <b>0.583</b>    | <b>0.313</b>       | <b>19.24</b>      | <b>0.570</b>    | <b>0.336</b>       |



Figure 6: **Image-to-3DGS Generation.** Five rendered views sampled from 3DGS scenes generated by our method.

## 6 EXPERIMENTS

### 6.1 EXPERIMENTAL SETUP

**Datasets.** Training uses no existing multi-view dataset; instead, we assemble a purpose-built corpus, denoted the *Lyra* dataset, in which the video model alone supervises the 3DGS decoder. The 3D reconstruction configuration comprises 59,031 images and the 4D configuration 7,378 videos. All instances derive from diverse text prompts spanning indoor and outdoor environments, humans, animals, and both photorealistic and imaginative content. We synthesize six camera trajectories per image (3D) or video (4D), producing 354,186 videos for 3D and 44,268 videos for 4D.

**Baselines.** We benchmark against state-of-the-art methods for generative single-image-to-3D reconstruction, namely ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Since *no source code is available* for evaluating these methods on our out-of-distribution evaluation set, we rely primarily on the quantitative results reported in the respective papers.

**Evaluation.** Consistent with prior work, we evaluate on single-image-to-3D using RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks and Temples (Knapitsch et al., 2017), adopting the evaluation protocol of Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b). Performance is quantified with the standard reconstruction metrics PSNR, SSIM, and LPIPS.

### 6.2 MAIN RESULTS

**Quantitative results.** Tab. 1 reports the single-image-to-3D evaluation. Our method surpasses all prior work on every benchmark and every metric. It follows that the principal avenue for further quality gains is a stronger video generative backbone, since our reconstructions inherit its fidelity directly. Additional quantitative comparisons on the *Lyra* dataset are provided in Appendix C.



Figure 7: **Ablations.** Each ablation is visualized by rendering an identical extreme novel viewpoint following image-to-3DGS generation; depth visualizations and additional ablations are provided in Appendix C.4.

**Qualitative results.** Novel-view renderings from our generated 3DGS scenes are shown in Fig. 1 and Fig. 6; video results will be released upon acceptance. Our method synthesizes high-fidelity novel-view content for 3D/4D scenes in unobserved regions while remaining consistent with the input image/video. Qualitative comparisons on the *Lyra* dataset are provided in Appendix C.

### 6.3 ABLATIONS

Tab. 2 substantiates our design choices by ablating individual components on diverse out-of-distribution prompts. Each variant is assessed by comparing its 3DGS renderings against the videos produced by our camera-controlled video diffusion model.

**Real data only.** In lieu of self-distillation, we train a model exclusively on real multi-view datasets, namely RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b). Although these datasets are standard in prior work, they yield limited generalization to out-of-distribution scenes.

**Self-distillation + real data.** Perhaps counterintuitively, joint training on self-distilled and real data yields no improvement over self-distillation alone. This indicates that the self-distilled supervision is sufficiently diverse and consistent to train a reconstruction model on its own.

**No depth loss.** The depth objective suppresses near-planar geometry and additionally produces a slight gain in image-space metrics. Depth maps are visualized in Appendix C.4.

**No opacity pruning.** Retaining all Gaussians in place of opacity-based pruning is evaluated here. Learning to prune the majority of Gaussians yields a more compact representation and a marginal improvement in visual quality. At  $H = 704$  and  $W = 1280$ , rendering requires 18 ms with pruning versus 30 ms without, a  $1.67\times$  speedup.

**No LPIPS loss.** Incorporating the LPIPS objective increases robustness to input inconsistencies and improves the preservation of high-frequency detail.

**No multi-view fusion.** A naive baseline regresses 3D Gaussians independently per camera trajectory and subsequently merges them into a single point cloud. We instead find a substantial benefit in learning the multi-view fusion across trajectories within our reconstruction blocks, in which each token attends to all others.

**No Mamba-2.** Substituting Transformer-only blocks for the joint Transformer/Mamba-2 blocks incurs a slight quality degradation. A single forward pass with  $V = 6$ ,  $L = 121$ ,  $H = 704$ , and  $W = 1280$  takes 3213 ms with the joint blocks versus 20922 ms with Transformer-only blocks, a  $6.5\times$  speedup.

**No latent-based 3DGS.** Pixel-space methods such as BTimer (Liang et al., 2025b) are limited to 12 input frames, whereas our latent-space formulation ingests up to 726. Accordingly, operating the 3DGS decoder in pixel space rather than latent space exhausts GPU memory.

Table 2: **Component ablation on the *Lyra* dataset.**

| Method                    | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ |
|---------------------------|-----------------|-----------------|--------------------|
| Ours                      | 24.77           | 0.837           | 0.224              |
| <b>Data</b>               |                 |                 |                    |
| real data only            | 19.08           | 0.659           | 0.413              |
| self-distill. + real data | 24.74           | 0.823           | 0.236              |
| <b>Loss</b>               |                 |                 |                    |
| w/o depth loss            | 24.31           | 0.811           | 0.247              |
| w/o opacity pruning       | 24.55           | 0.820           | 0.237              |
| w/o LPIPS loss            | 23.74           | 0.766           | 0.370              |
| <b>Architecture</b>       |                 |                 |                    |
| w/o multi-view fusion     | 17.73           | 0.632           | 0.446              |
| w/o Mamba-2               | 24.58           | 0.818           | 0.241              |
| w/o latent 3DGS           |                 | OOM             |                    |

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## 7 CONCLUSION

We have presented *Lyra*, a framework for 3D and 4D generation conditioned on a single image or video. Rather than collecting multi-view datasets, we cast camera-controlled video diffusion models as teachers within a self-distillation framework that trains a student 3DGS decoder. The decoder operates in the latent space of the video model and reconstructs 3D Gaussians directly, without post-processing or optimization. This design improves generalization and viewpoint coverage across diverse scenes, and the dynamic extension establishes the feasibility of 4D generation from monocular video. At present, the scale and consistency of the generated scenes are upper-bounded by the capacity of the underlying camera-controlled video diffusion model; consequently, stronger video diffusion backbones will directly enable large-scale, consistent scene synthesis. A further promising direction is to integrate auto-regressive techniques (Chen et al., 2024a) into the framework to support large-scale generation. Finally, explicitly modeling motion and tracking within the reconstruction network, as in concurrent work (Lin et al., 2025), may improve the fidelity of synthesized motion.

## 8 ETHICS STATEMENT

As with all generative-AI technologies, the proposed method carries a risk of misuse, including the fabrication of misleading or synthetic 3D/4D content. Notwithstanding these risks, the objective of this research is to advance simulation, robotics, and embodied AI by providing scalable and controllable tools for data generation. We anticipate that the principal contributions of this work materialize through the following capabilities and insights.

- **Realistic simulation for embodied agents:** The method affords controllable and physically consistent camera motion within generated 3D and 4D environments, thereby supporting the training and evaluation of agents that must perceive, navigate, and interact with complex scenes.
- **Improved scalability for data engines:** By eliminating multi-view capture and costly per-scene optimization, the method synthesizes high-fidelity, customizable scenes at scale, a property well suited to research in robotics, reinforcement learning, and closed-loop simulation.
- **Improved technical understanding:** As an academic study, this work provides a foundational case study of the interaction between camera control and spatiotemporal generative models, of direct relevance to research in computer vision, graphics, and embodied AI.

While we caution against deceptive or otherwise wrongful use of this technology—wherein misleading or unverifiable media could be produced from realistic generative outputs—we hold that the adoption of appropriate safeguards (e.g., provenance tracking, dataset documentation, and rigorous evaluation practices) will help ensure that generative models advance science and society constructively.

## 9 REPRODUCIBILITY STATEMENT

To enable full reproduction of our results and to accelerate subsequent research, we will release the training and inference code, model weights, and data for both 3D and 4D generation upon acceptance.

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## A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section supplies additional detail on the GEN3C (Ren et al., 2025) video diffusion backbone introduced in Sec. 3.1 and describes our procedure for further improving its 3D consistency.

**Conservative mask refinement.** A salient limitation of GEN3C’s forward warping is background leakage: object parts occluded in the source view (e.g., the lion body in Fig. 8) become visible under novel viewpoints yet are not flagged in the disocclusion mask. The video diffusion model consequently produces incomplete foreground completion, as shown in Fig. 8 (second and fourth columns).

To rectify this, we adopt a conservative masking strategy: because the geometry of occluded regions in  $\mathbf{P}^{t,v}$  is unknown, we treat all such regions as occupied and derive refined disocclusion masks accordingly. This lets the video model reason about potentially visible foreground regions overlooked by standard point-based rendering.

Concretely, we construct a triangular mesh from the camera-space points  $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$  by connecting spatially adjacent pixels, following Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where  $\mathbf{p}_{u,v}$  denotes the 3D point at pixel  $(u, v)$  in  $\mathbf{P}_c^{t,v}$ . This surface mesh acts as a boundary separating visible from invisible regions. For each rendered pixel, we compare the standard point-based depth  $\mathbf{D}^{t,v}$  against the mesh-interpolated depth  $\mathbf{D}_{\mathcal{M}}^{t,v}$  obtained by ray-surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where  $\mathbf{M}_{\text{orig}}^{t,v}$  is the original disocclusion mask from forward warping and  $\epsilon$  is a small tolerance. Whenever the mesh surface lies closer than the original points, we conservatively mask the corresponding regions as candidate foreground disocclusions. This yields more reliable guidance for video generation, particularly under large viewpoint changes, as shown in Fig. 8 (third and fifth columns).

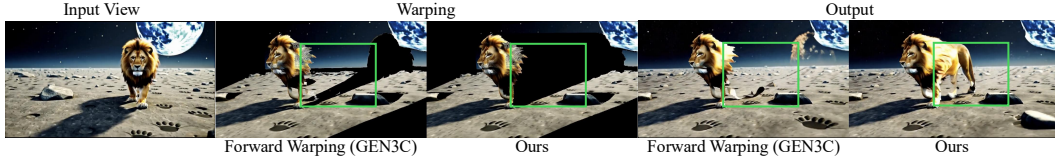


Figure 8: **Rendering function comparison.** Our model builds upon GEN3C (Ren et al., 2025), which conditions camera control on forward-warped images. We contrast forward warping (left) with our conservative rendering function (right). In forward warping the highlighted region is erroneously filled with background pixels, whereas our formulation masks it correctly, enabling proper completion.

## B MODEL DETAILS

**Progressive training.** Training the 3DGS decoder directly on 6 trajectories of 121 frames at  $704 \times 1280$  is computationally prohibitive; we therefore adopt a progressive training schedule, reported in Tab. 3. The final static pre-trained model additionally serves to initialize the dynamic model. Total training consumes 6 days on 8 NVIDIA A100 80 GB GPUs. We use gsplat (Ye et al., 2025) as the 3DGS implementation.

**Subsampling Gaussians.** The count of pixel-aligned 3D Gaussians scales linearly in the spatial and temporal resolution. Our generated scenes span 726 frames at  $704 \times 1280$  resolution, corresponding to 654,213,120 per-pixel Gaussians. To control this, we emit a single 3D Gaussian per  $8 \times 8$  spatial neighborhood, reducing the total by a factor of 64 to 10,222,080. Opacity-based pruning further reduces the count to 2,044,416, yielding a substantially more compact representation.

**Joint Transformer-Mamba blocks.** We adopt the reconstruction block of Long-LRM (Ziwen et al., 2024), in which each block consists of one Transformer layer followed by seven Mamba-2 layers; the

Table 3: **Progressive training schedule.** Configuration of each successive training stage.

| Stage   | $H \times W$ | $L$ | $V$ | $S$ | $B$ | Steps |
|---------|--------------|-----|-----|-----|-----|-------|
| Static  |              |     |     |     |     |       |
| 1       | 176×320      | 17  | 1   | 17  | 4   | 10k   |
| 2       | 176×320      | 49  | 1   | 49  | 4   | 2.5k  |
| 3       | 352×640      | 49  | 1   | 49  | 2   | 2.5k  |
| 4       | 704×1280     | 49  | 1   | 49  | 1   | 2.5k  |
| 5       | 704×1280     | 121 | 1   | 9   | 1   | 57.5k |
| 6       | 704×1280     | 121 | 1-6 | 9   | 1   | 7k    |
| Dynamic |              |     |     |     |     |       |
| 7       | 704×1280     | 121 | 6   | 12  | 1   | 10k   |

block is instantiated twice for a total of 16 layers with a hidden width of 512. Notably, our network is considerably smaller than those of Long-LRM (Ziwen et al., 2024) and Wonderland (Liang et al., 2025a), which employ 24 layers with a hidden width of 1024. Although pure Transformers are generally more expressive, they are markedly slower in both training and inference than Mamba-2. As both configurations were trained under an identical compute budget, we conjecture that Transformer-only blocks would require substantially more training iterations to match the rendering quality.

**Camera encoding.** Following prior work (Kant et al., 2024; Zhang et al., 2024e), cameras are encoded as Plücker embeddings. The embeddings are computed at full spatial and temporal resolution and then projected into the video latent space by the pre-trained RGB encoder. Because the RGB encoder accepts 3 input channels whereas Plücker embeddings are 6-dimensional, we encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins separately, then concatenate the resulting latent encodings along the channel dimension.

**Time encoding.** Analogously, to render the 1-dimensional time input compatible with the 3-channel input expected by the RGB encoder, we concatenate a 2-dimensional sinusoidal embedding to the scalar time value along the channel dimension. These values are then replicated across the spatial dimensions prior to encoding the time into the video latent space. This scheme permits the source and target times to be encoded independently into the compressed latent representation.

**Dynamic data augmentation.** Fig. 9 illustrates the dynamic data augmentation procedure for two representative camera trajectories. The procedure yields pairs of supervision views that observe the scene at an identical motion state from opposing extreme viewpoints.

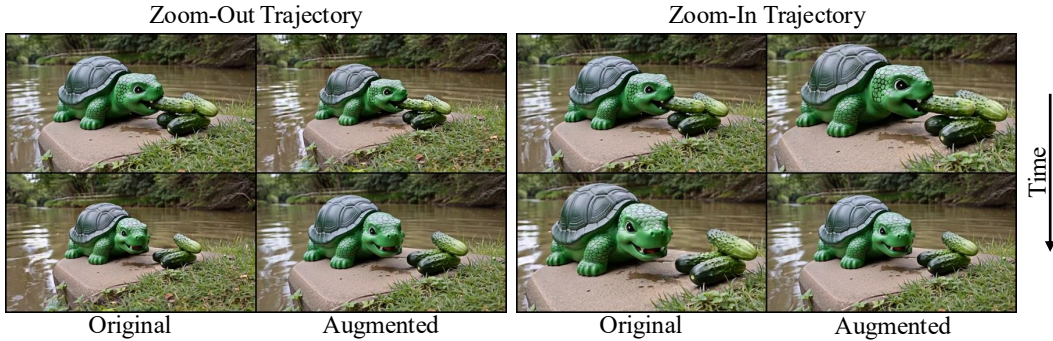


Figure 9: **Dynamic data augmentation videos.** We augment the supervision data with a motion-reversed video so that each timestep is observed under full spatial coverage, thereby eliminating low-opacity artifacts at early timesteps. We show two representative trajectories, zoom-out and zoom-in, together with their corresponding augmented videos; the augmented videos reverse the original camera motion.



Figure 10: **Image-to-3DGS Comparison.** We compare our method against BTimer (GEN3C), showing five rendered views per generated scene. Our method exhibits markedly fewer artifacts and higher fidelity.

## C ADDITIONAL EVALUATION

### C.1 ADDITIONAL BASELINE DETAILS

We further compare against BTimer (Liang et al., 2025b), a recent method for joint 3D and 4D reconstruction from posed images or videos. At our request, the BTimer authors conducted experiments on the same scenes used for *Lyra*. As BTimer is purely regression-based, we couple it with our GEN3C (Ren et al., 2025) video diffusion backbone: we supply the identical camera trajectories used by our model and feed the RGB-decoded GEN3C videos as input to BTimer. BTimer operates in pixel space and requires 12 input images. As noted in the main paper, pixel-space models such as BTimer exhaust GPU memory when ingesting the 726 high-resolution frames generated by GEN3C directly. We therefore uniformly subsample the GEN3C frames. For static evaluations, we select 12 frames that uniformly cover the available viewpoints. For dynamic evaluations, we sample 11 frames spanning the viewpoint range and additionally include the bullet-time frame corresponding to the reconstructed motion, guaranteeing that the target motion state is always present as a reference. We denote this integrated baseline BTimer (GEN3C), i.e., BTimer combined with GEN3C.

### C.2 ADDITIONAL STATIC 3D EVALUATION

For image-to-3DGS generation, Fig. 10 presents qualitative comparisons with BTimer (GEN3C) and Tab. 4a reports the corresponding quantitative results. Our method attains higher fidelity with fewer artifacts.

### C.3 DYNAMIC 3D EVALUATION

**BTimer (GEN3C) comparison.** For dynamic 3D evaluation, we compare against BTimer (GEN3C), as described in Sec. C.1, on 100 out-of-distribution videos drawn from our dynamic *Lyra* dataset. To ensure a fair comparison, videos are center-cropped to  $512 \times 512$ , the resolution at which BTimer was predominantly trained. The results in Tab. 4b show that our method substantially outperforms BTimer (GEN3C).

Table 4: Quantitative results on the static and dynamic *Lyra* datasets.

| Method         | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ | Method         | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ |
|----------------|-----------------|-----------------|--------------------|----------------|-----------------|-----------------|--------------------|
| BTimer (GEN3C) | 16.32           | 0.580           | 0.427              | BTimer (GEN3C) | 20.29           | 0.687           | 0.315              |
| Ours           | <b>24.92</b>    | <b>0.834</b>    | <b>0.183</b>       | Ours           | <b>23.07</b>    | <b>0.779</b>    | <b>0.231</b>       |

(a) Comparison on the static *Lyra* dataset.

(b) Comparison on the dynamic *Lyra* dataset.

#### C.4 DEPTH VISUALIZATION

Fig. 11 visualizes the depth maps derived from our generated 3DGS. Depth supervision suppresses near-planar geometry without any degradation in visual quality.

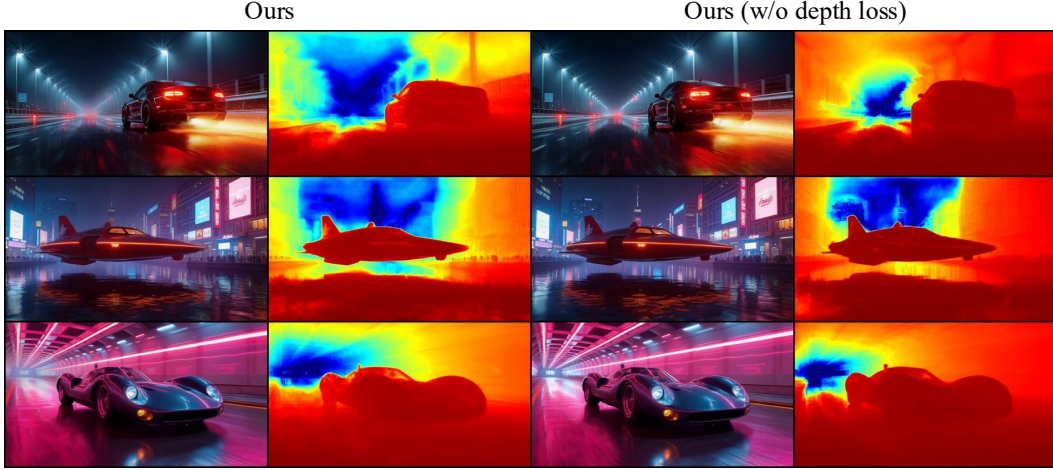


Figure 11: **Depth loss ablation.** We show 3DGS renderings alongside their corresponding depth maps. Employing the depth loss as auxiliary supervision suppresses near-planar geometry without any loss of visual quality.

#### D ROBOT SIMULATION

A primary motivation for constructing a diverse 3D generator is its capacity to synthesize simulation environments for autonomous agents. To this end, we implement an end-to-end pipeline: our model first generates 3D scenes from text, the 3D Gaussians are exported to a .ply file, and the .ply file is converted to the .usdz format via the export function of 3DGUT (Wu et al., 2025a). The resulting .usdz asset is then imported into the NVIDIA Isaac robot development platform as a physically based virtual environment, within which AI-based robots can be trained and evaluated under diverse conditions. Early demo results will be released upon acceptance; a screenshot of the interface is provided in Fig. 12.

#### E RELATED WORK

This section reviews further related work. The distinctions between our approach and the prior methods CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b) are summarized in Fig. 13.

**3D generation.** Early 3D generation methods were largely restricted to single object categories, extending GANs into the 3D domain by employing neural renderers as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-EI et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). The introduction of CLIP-based supervision (Radford et al., 2021) enabled more flexible and diverse asset creation, supporting both text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). More recently, diffusion models have substantially improved quality by supplanting CLIP guidance with Score Distillation

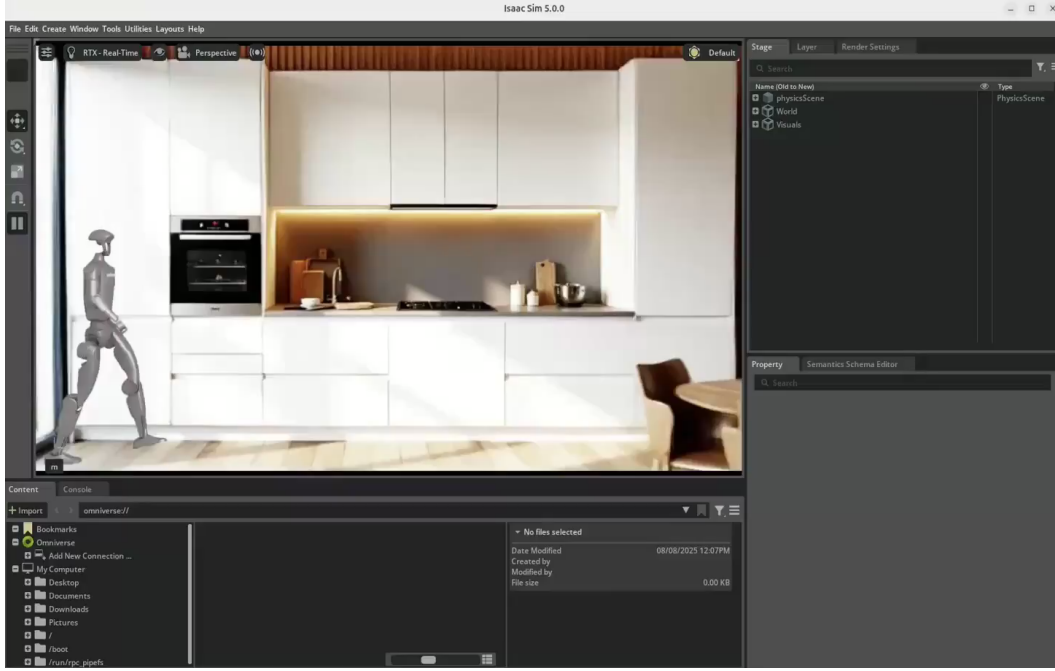


Figure 12: **Robot Simulation.** A single frame of a robot simulation executed in the Isaac Sim 5.0 framework, taking a 3DGS scene generated by our method as input.

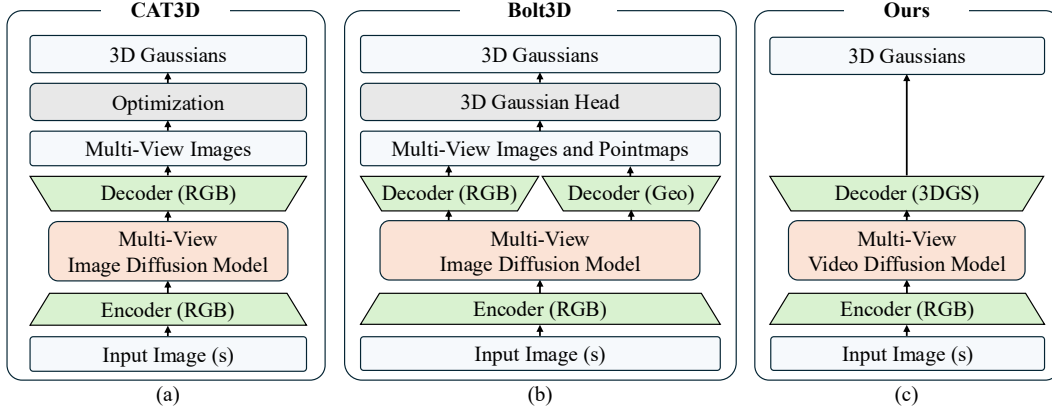


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) introduces a multi-view image diffusion model that generates images from novel viewpoints, which are subsequently reconstructed into 3D Gaussians via optimization. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to output pointmaps through a geometry autoencoder and, in place of optimization, predicts the 3D scene with a feed-forward 3D Gaussian head. In contrast, our method builds upon a pre-trained camera-controlled video diffusion model and *directly* decodes the multi-view video latents into 3D Gaussians.

Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al., 2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

To improve structural coherence, several methods enforce multi-view consistency by generating scenes from distinct viewpoints (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), whereas others cast scene synthesis as iterative inpainting (Höllein et al., 2023; Shriram et al., 2024). A further line of work lifts 2D images into 3D using NeRF (Mildenhall et al., 2020), 3D Gaussian

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Splatting (Kerbl et al., 2023), or mesh-based representations coupled with diffusion models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

**Feed-forward 3D models.** A complementary body of work develops fast, feed-forward methods that directly regress 3D content from image or text inputs (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e;d; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). In contrast to our method, however, these approaches are confined to static 3D scenes. Others specialize to particular scene classes such as faces (Kirschstein et al., 2025). Concurrent work (Liang et al., 2025b; Xu et al., 2025) addresses real scenes but cannot handle diverse generated scenes or large viewpoint changes.

**4D generation.** Recent years have witnessed rapid progress in 4D generation, i.e., dynamic 3D scene synthesis. Many methods condition on input text prompts or images. Following the introduction of large-scale generative models for this task (Singer et al., 2023), subsequent work has advanced both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

While text conditioning is widely adopted, alternative methods lift 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Related efforts incorporate physics priors into the generation pipeline (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025) or improve motion controllability via template-driven formulations (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b). A parallel direction targets compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

A further strand extends 3D GAN frameworks to the 4D setting by training directly on 2D video (Bahmani et al., 2023a; Xu et al., 2023). Such methods are typically constrained by small-scale, single-category datasets that limit generalization, and most remain object-centric, neglecting background structure. Consequently, their visual fidelity falls short of the photorealism attained by recent video generation models, which our method exploits.

**Camera-conditioned video models.** Recent progress has centered on introducing camera control into video diffusion models. The pioneering MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent methods (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) strengthened the conditioning by encoding cameras as Plücker coordinates, whereas others follow MotionCtrl in simply flattening the poses (Wang et al., 2025b). A distinct line of work (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) enables camera motion control without any additional trainable parameters. Notwithstanding these advances, all of the above rely on U-Net-based architectures. Other approaches adopt low-rank parameterizations for camera control (Zhang et al., 2025b).

Several recent methods continue to extend U-Net-based camera control (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), while others begin to explore diffusion transformers. Cheong et al. (Cheong et al., 2024) propose one such transformer-based framework, albeit with limited scene and visual quality. DimensionX (Sun et al., 2024d) introduces joint space-time control in diffusion transformers but depends on pre-defined, non-continuous camera trajectories. Other works (Huang et al., 2024a; Mai et al., 2025; Jiang et al., 2025b) instead estimate pose and geometry with a video DiT. CAT4D (Wu et al., 2025b) derives a multi-view video diffusion model by fine-tuning a multi-view backbone. Large-scale synthetic data generation further improves performance, as demonstrated by SynCamMaster (Bai et al., 2025b) and

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1512 ReCamMaster (Bai et al., 2025a), which achieve strong camera-controlled video generation with  
1513 transformer-based architectures. Finally, 4DiM (Watson et al., 2024) trains a space-time diffusion  
1514 model from scratch for single-image novel view synthesis. A separate line of work edits the camera  
1515 motion of videos with underlying scene motion via camera retargeting (Seo et al., 2025; Ma et al.,  
1516 2025). A prominent direction conditions camera-controlled video generation on a spatiotemporal  
1517 cache, e.g., point clouds or optical flow (Wu et al., 2025c; Ren et al., 2025; Wang et al., 2025d; Cao  
1518 et al., 2025; YU et al., 2025; Jin et al., 2025; Xing et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu  
1519 et al., 2025).